

BEATRICE LOWMAN

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Brain-computer interfaces · Neural decoding · Computer vision for neuroimaging · Clinical neurotechnology

EDUCATION

University of California, Berkeley

Expected May 2026

B.A. in Cognitive Science | B.A. in Data Science

Major GPA: 3.83 | Cumulative GPA: 3.7

Relevant Coursework: Cognitive Neuroscience; Brain, Mind, and Behavior; Computational Models of Cognition; Natural Language Processing; Data Engineering; Intro to Deep Learning; Concepts of Probability

RESEARCH EXPERIENCE

UC Berkeley CALM Lab

Sept 2024 – June 2025

Research Assistant & Computer Programming Specialist

Berkeley, CA

- Served as Research Assistant and Computer Programming Specialist on the N-ACT (Neurobehavioral Affective Control Training) randomized controlled trial — a novel 8-week cognitive remediation intervention for emotion-related impulsivity and rumination — contributing to both the development and evaluation of the program under Dr. Sheri Johnson and Dr. JD Allen.
- Conducted participant sessions with a clinical population under IRB protocol, administering laboratory cognitive tasks, GSR and eye-tracking data collection, self-report measures, and EMA surveys; maintained detailed session documentation and records.
- Built and optimized multiple computerized cognitive training tasks for the intervention, including the adaptive Emotional Stop-Signal Task (ESST) and additional hot and cold cognition paradigms.
- Developed presentation materials for the B.A.A.S. conference showcasing experimental design and progress.

PROJECTS

Brain-to-Text: Optimizing Neural Speech Decoding from iEEG

Fall 2025

Neurotech@Berkeley

- Built upon the Brain-to-Text Benchmark '24 to optimize a neural decoding pipeline translating intracranial EEG (iEEG) signals from an ALS patient with tetraparesis into intended sentences.
- Baseline pipeline used an RNN-GRU to compute phoneme probabilities, fed into an N-Gram and LLM rescoring stage (baseline WER: 5.77%); led a team implementing multiple optimizations to reduce word error rate.
- Optimizations included: Temporal Masking (frequency and random) for noise robustness; diphone-based lexicon; rebuilt CTC decoding pipeline (beam search + lexicon + beam search); RNNT + RNNT loss; Cold Fusion LLM integrating phoneme CTC embeddings with LLaMA 3.3 hidden states.
- Developed and presented academic-style poster at the Neurotech@Berkeley Symposium, Fall 2025.

Early Dementia Detection via Hybrid CNN-ViT Model

Spring 2026

Neurotech@Berkeley

- Leading a project team fine-tuning MedViT-2 — a hybrid CNN-Transformer architecture — for four-class dementia stage classification from neuroimaging data, the first evaluation of MedViT on neuroimaging or dementia-related tasks.
- Model achieved class-specific F1 scores exceeding 99% across all stages with overall accuracy above 99%, substantially outperforming existing baselines on the same dataset.
- Results to be presented at the California Neurotechnology Conference (C.T.N.C.), April 2026.

3D MRI Brain Tumor Segmentation

Fall 2024 – Spring 2025

Neurotech@Berkeley

- Developed a U-Net convolutional neural network to segment brain tumors in 3D MRI scans, trained on 369 pre-operative volumes across four modalities (T1, T1Gd, T2, FLAIR) from the BraTS dataset.
- Iterated on model architecture — progressing from a standard U-Net with Dice Loss to an attention-gated U-Net with Focal Tversky Loss to address class imbalance across necrotic tissue, edema, and enhancing tumor regions.
- Presented at the California Neurotechnology Conference (C.T.N.C.), 2025.

Music Reconstruction from iEEG

Fall 2025

Neurotech@Berkeley

- Led a project team replicating the Knight Lab (UC Berkeley) approach to reconstructing music from iEEG recordings, implementing linear encoding/decoding models linking auditory spectrogram features with high-frequency brain activity.
- Team conducted outreach with Ludovic Bellier, first author of the source paper, who provided methodological guidance on the replication.

EEG-Guided Meditation & AI-Generated Artwork

2023

Neurotech@Berkeley

- Published in Neurotech@Berkeley's MIND Magazine on EEG-guided meditation and its applications.
- Featured on the Neurotalk Spotify Podcast reviewing a project on EEG and AI-generated artwork.

INDUSTRY EXPERIENCE

Google DeepMind – AI UX Team

Sept 2025 – March 2026

Project Manager (Contract)

Berkeley, CA

- Built and prototyped a framework for trust-adaptive AI systems that dynamically adjust autonomy based on real-time user behavior patterns.
- Designed a closed-loop system using contextual bandits to continuously optimize human-AI interaction for maximized user comfort.
- Developed a natural language calendar agent for intelligent scheduling support and conflict resolution.

Xiberline Inc.

Neural Engineering Intern

Jan 2025 – Aug 2025

Remote

- Implemented back-end software for a novel non-invasive BCI prototype designed to enhance emotional wellbeing and cognition.
- Developed a real-time pipeline for facial landmark extraction and processing using MediaPipe and OpenCV.
- Extracted interpretable features from high-dimensional EEG data using PCA, ICA, and facial action coding.
- Trained ML models (LSTM, SVM) to predict EEG patterns; evaluated and optimized model performance.

CONFERENCE PRESENTATIONS

Marquez-Garcia, M., Rosenberg, L., Werbach, P., Carrillo, J., Kapoor, G., Lowman, B., Sharma, A., Glantz, S., Bangser, S., Liu, V., Atkinson, O., & Johnson, S. *Hot Thoughts, Smart Moves: Neurobehavioral Training for Rumination and Impulsivity*. Poster presentation at the Bay Area Affective Science Symposium (B.A.A.S.), 2025.

Lowman, B., Sandoval, E., Yang, J., Gricelda Sotelo, S., Parker, N., & Billing, J. *Computer Vision for Brain Tumor Detection*. Poster presentation at the California Neurotechnology Conference (C.T.N.C.), 2025.

Lowman, B., Gu, N., Billing, J., Singh, K., & Hadveck, R. *Improving Dementia Stage Classification with Hybrid CNN-ViT Model*. Poster presentation at C.T.N.C., 2026. (Forthcoming, April 26, 2026)

VOLUNTEER EXPERIENCE

ACC Senior Services

Jul 2020 – Nov 2020; Dec 2025 – Feb 2026

U.S. Census Intern; Data Volunteer

Sacramento, CA

- Worked directly with senior community members to provide individualized assistance with Census completion as part of an inclusion initiative for underrepresented populations.
- Used Excel and Python to collect, analyze, and present data on undercounted zones and populations to guide targeted outreach efforts.
- 2025–2026: Streamlined participant and volunteer matching system and reorganized data tables to improve operational efficiency.

My Sister's House (Domestic Violence Shelter)

Jul 2018 – Aug 2022

Volunteer & Crisis Line Responder

Sacramento, CA

- Trained and certified advocate and crisis line responder supporting women escaping domestic violence.
- Provided direct support to clients in crisis, assessing needs and connecting them to appropriate services.
- Organized, staffed, and managed fundraisers including clothing drives, informational fairs, and art shows.

CERTIFICATIONS

CITI Program – Human Research: Social and Behavioral Research Investigators

Sept 2024

Collaborative Institutional Training Initiative (CITI) · University of California, Berkeley

SKILLS

Research & Clinical:	Human subjects research (IRB), GSR and Tobii eye-tracking data collection, cognitive task administration, REDCap, BioLAB, PsychoPy, Inquisit, E-Prime
Programming & ML:	Python, Java, SQL, MQL, deep learning, computer vision, transformers, LLMs
Data Science:	Data engineering, visualization, dimensionality reduction, ML pipelines, spreadsheets
Academic:	Academic writing, technical presentation, literature synthesis, conference speaking
Interpersonal:	Team leadership, active listening, conflict de-escalation, cultural sensitivity, problem solving